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SMART INDIA HACKATHON 2025

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TITLE PAGE

- **Problem Statement ID –25014**
- **Problem Statement Title-Waste Segregation Monitoring System for Urban Local Bodies**
- **Theme-Clean and Green Technology**
- **PS Category- Hardware**
- **Team ID-70782**
- **Team Name-InnovaMesh**

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Waste Segregation Monitoring System for Urban Local Bodies

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Problem Statement

Waste Segregation Monitoring System for Urban Local Bodies

Solution

IoT-enabled Segregation

Smart bin with IoT sensors and **mounted camera**.Identifies and **classifies waste types automatically**.Reduces manual effort in segregation

Shaking & Partition Design

Mechanical flaps and AI-based detection separate biodegradable, non-biodegradable, and hazardous waste, minimizing errors in disposal.

Smart Monitoring

Monitoring of waste levels supports planned collection schedules, preventing overflow and maintaining cleanliness.

Uniqueness

IoT + Visual Integration

By combining **sensors and a camera**, the bin achieves **accurate waste detection.**This improves sorting efficiency and **reduces errors** compared to normal bins.

User-Friendly QR Code

A **QR code** allows users to report issues quickly and connects them directly to a mobile app, **enabling faster maintenance responses.**

Source-Level Compliance

The system **directly addresses** improper household segregation, solving the problem before it reaches processing units.

WorkFlow

Waste Input

The user deposits waste into the bin, initiating the **automated segregation process**.

Image Capture

Camera captures the waste image.
Image is **processed for AI-based classification.**

AI Classification

The **AI model identifies the type of waste,** such as **biodegradable or non-biodegradable** and **harzards**, ensuring accurate sorting.

Biodegradable Waste

If the waste is detected as **biodegradable,** the system **directs it to the bio bin** for **eco-friendly disposal.**

Non-Biodegradable Waste

If the waste is **non-biodegradable**, the **system routes it for specialized handling.**

Bin Allocation

Servo-operated flaps **allocate the waste** into the correct bin, ensuring precise segregation.

Level Monitoring

Ultrasonic sensors monitor bin levels to **prevent overflow** and optimize collection schedules.

Cloud Update

Data from the bins is transmitted to a **cloud platform** such as a website or Firebase for **centralized storage and monitoring.**

Dashboard View

A dashboard **displays bin levels,alerts, and analytics**,provide actionable insights for **efficient waste management.**

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TECHNICAL APPROACH

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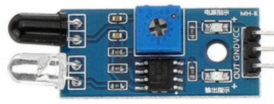
1.

The User is disposing waste (BIO/NON-BIO/HAZARDOUS)

2.



IR SENSOR- detect the object

3.

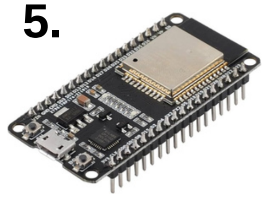
The lid of the bin will be opened automatically

This Automation of lid opening is done by SERVO MOTOR

4.

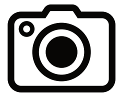
The garbage comes to First Level Segregation

5.



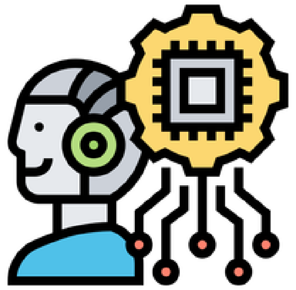
(Esp32-microprocessor)

(Esp Cam)



The ESP CAM module is used observe the types of garbage (BIO/NON-BIO/HAZARDOUS) and the ML data algorithm trained and fed into ESP

6.



The ML distinguish the input (Garbage)

The Circular Rotation of bin is done by SERVO MOTOR based on classifciation

7.

BIO Classification

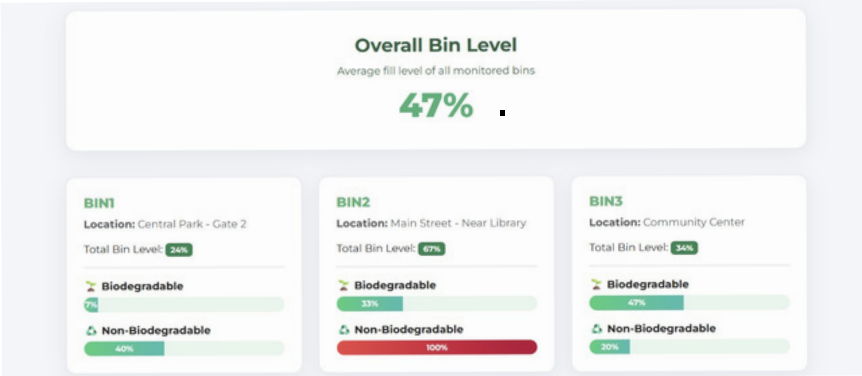
MOISTURE Sensor is used to measure the moisture content from the kitchen waste

NON-BIO Classification

INDUCTIVE PROXIMTY Sensor is used to measure the metal from the content

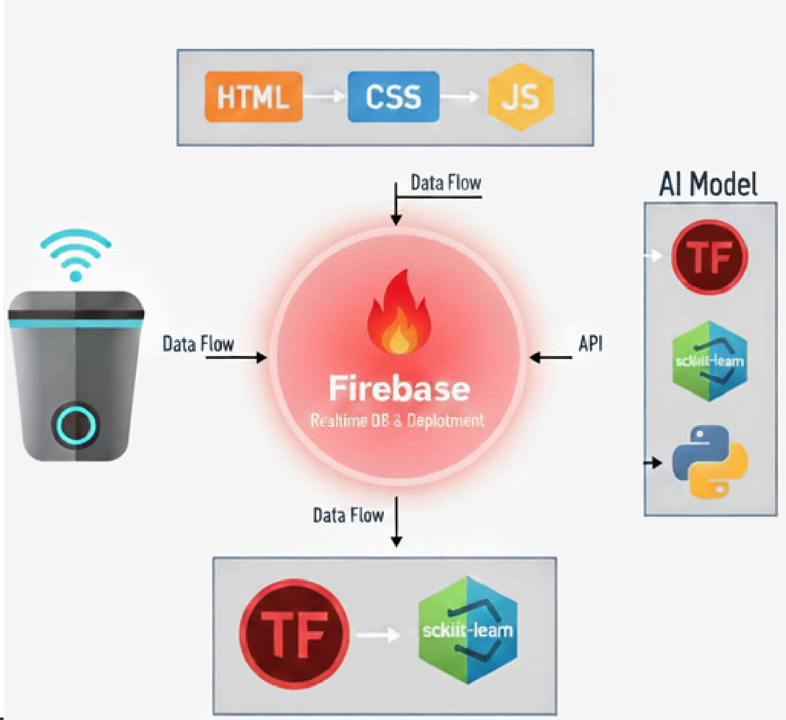
11.When the User is facing any trouble and wants direct connection with Authority

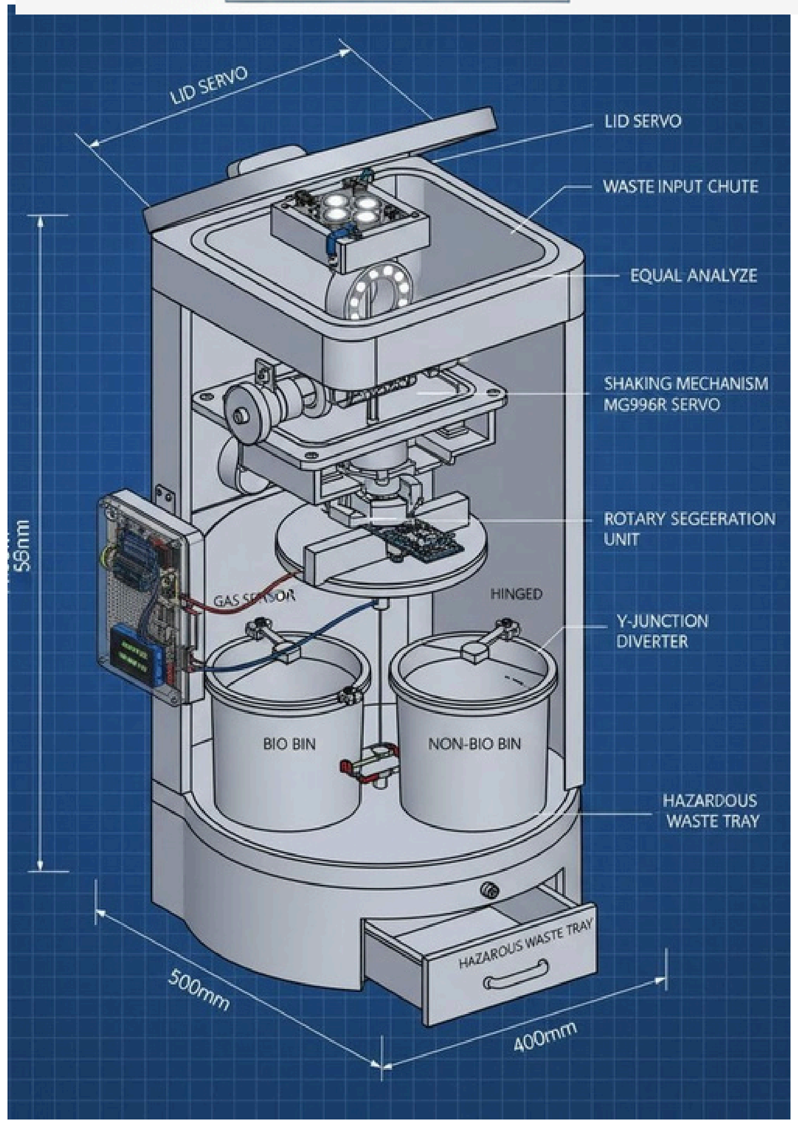
10. Sensor data is sent to Firebase, and the dashboard displays bin fill percentage.



9.When the bin is filled to or above 80%

The ULTRASONIC sensor is used to alert the user and authority about the filled bin





FEASIBILITY AND VIABILITY

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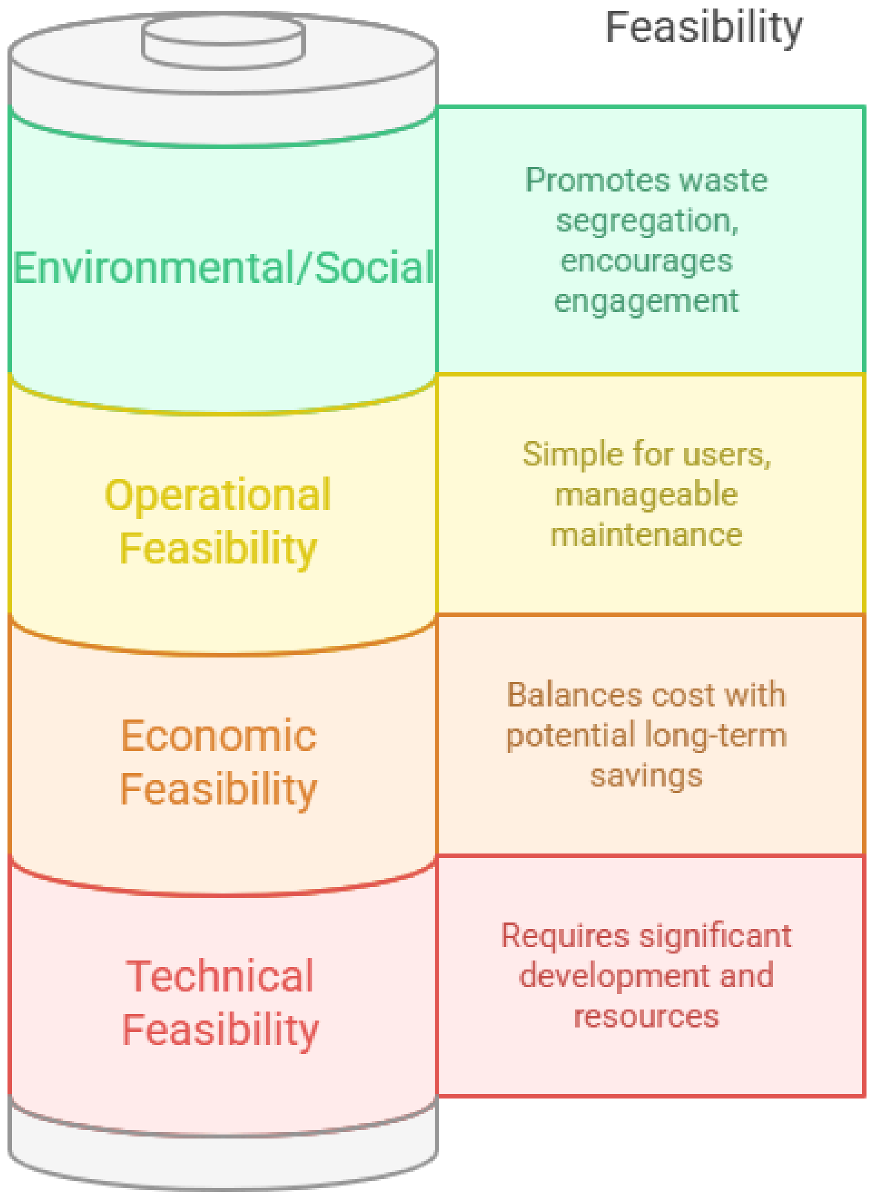
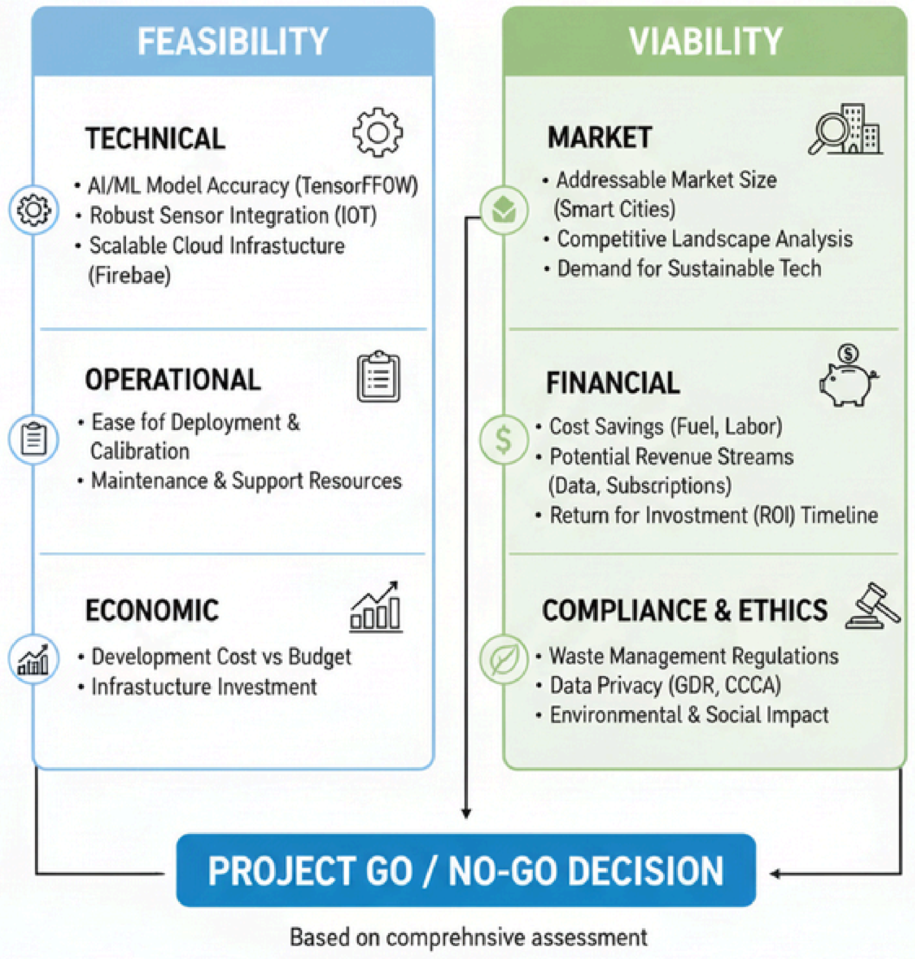
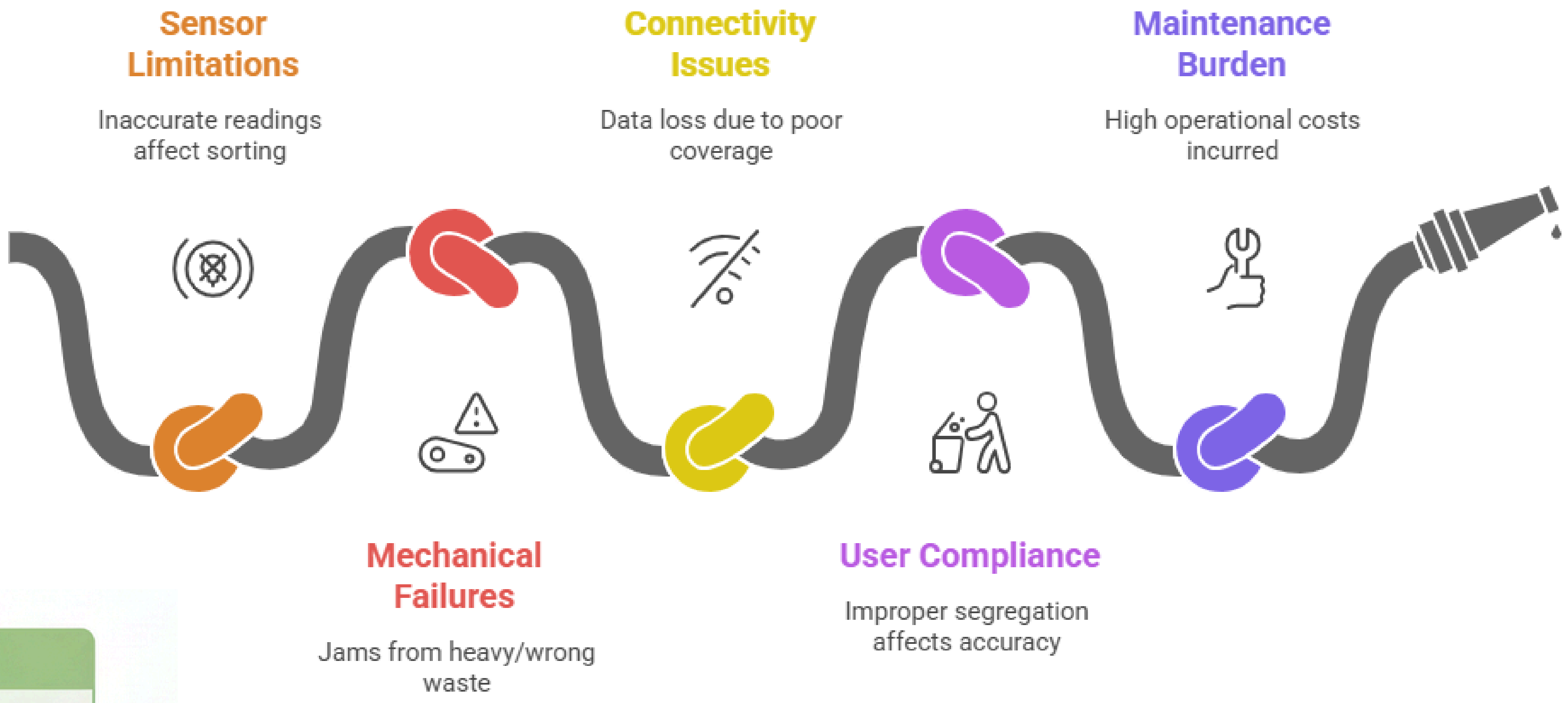
Feasibility spectrum from impossible to easily achievable goals



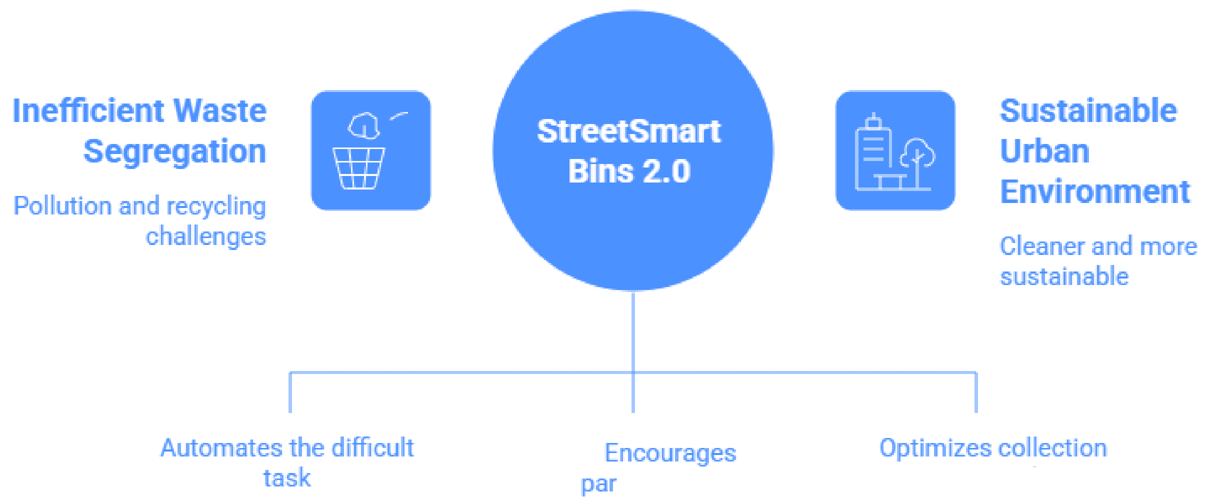
Made with Napkin



Smart Waste System Challenges



Automating Urban Waste Segregation



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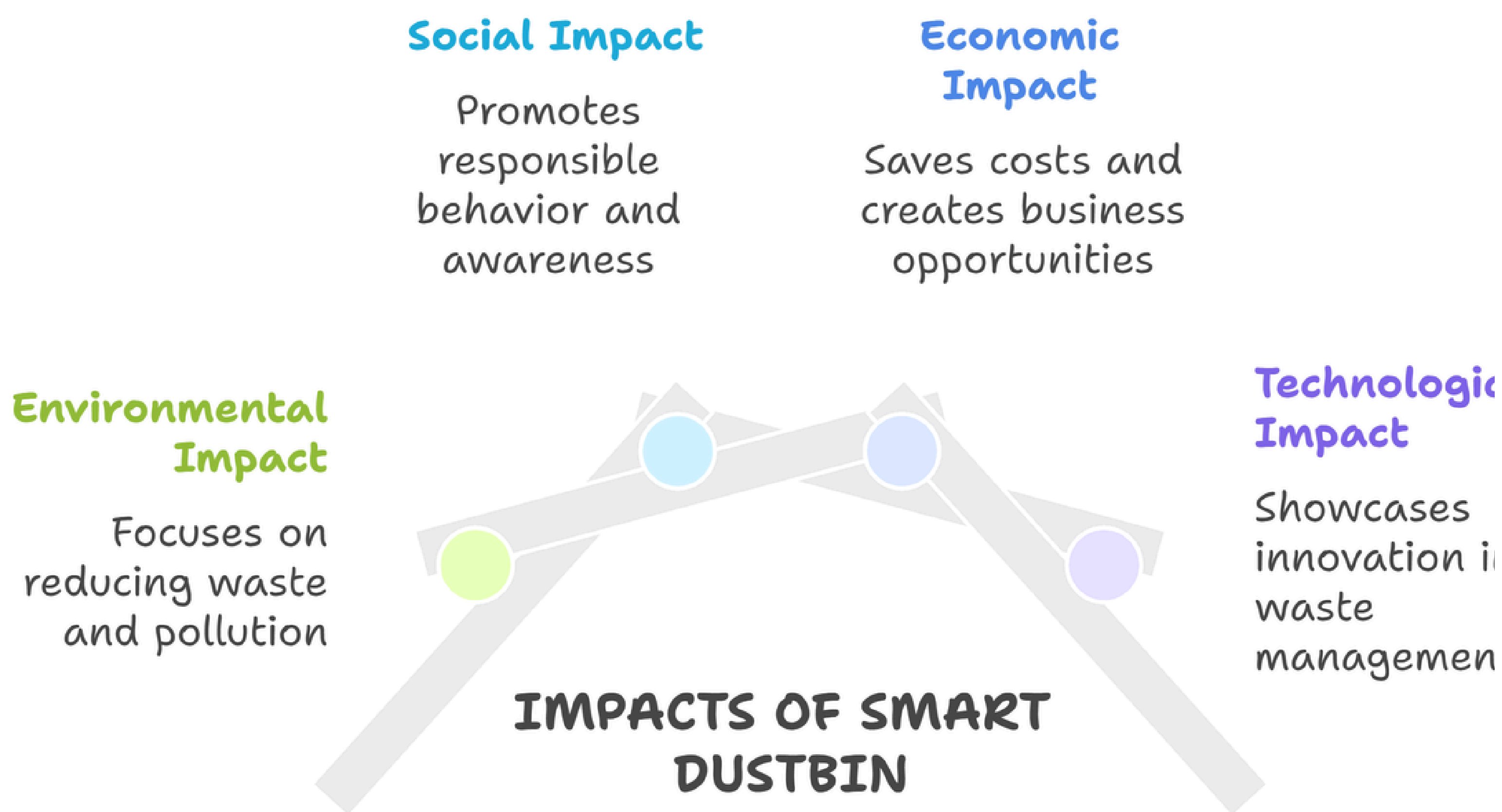
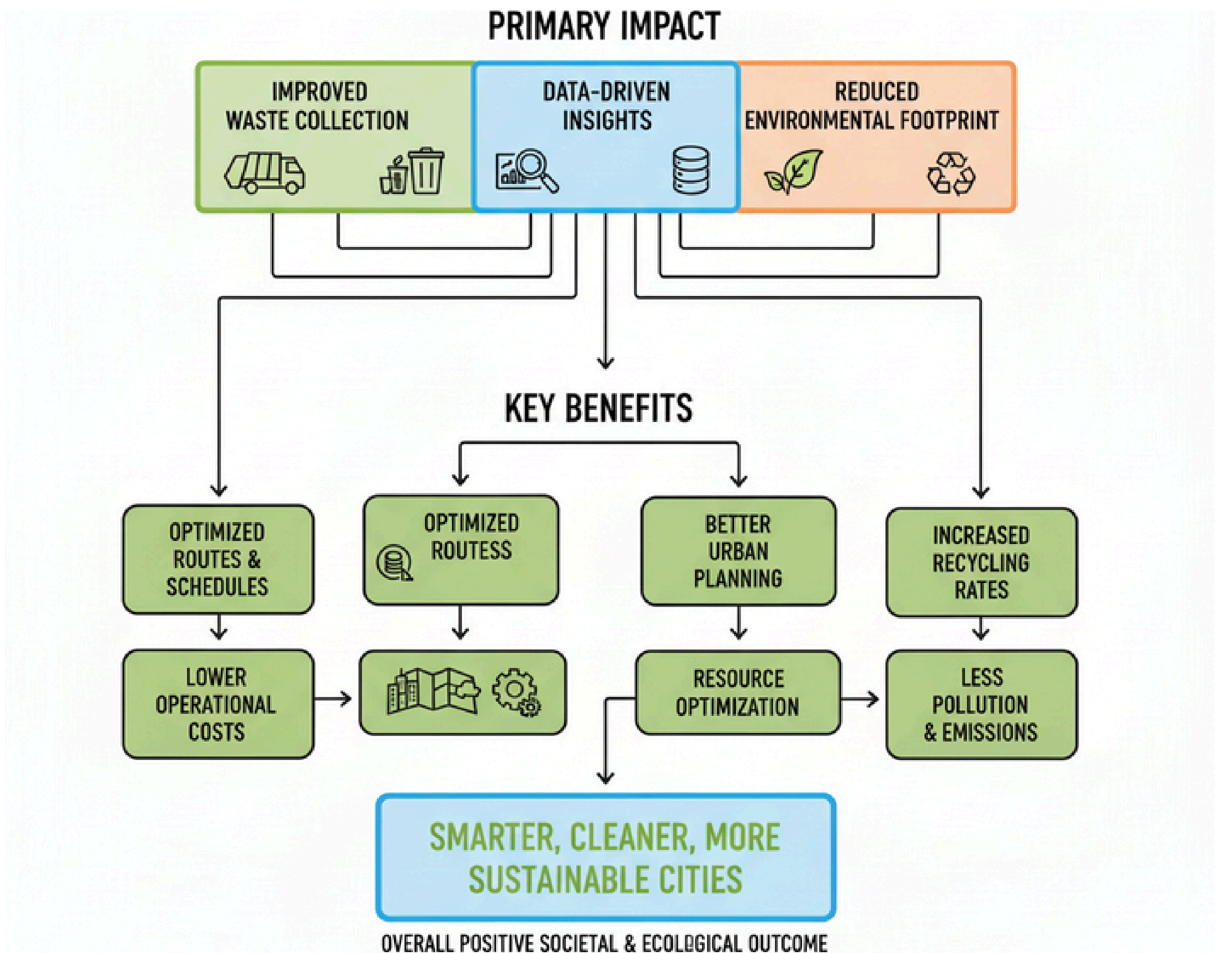
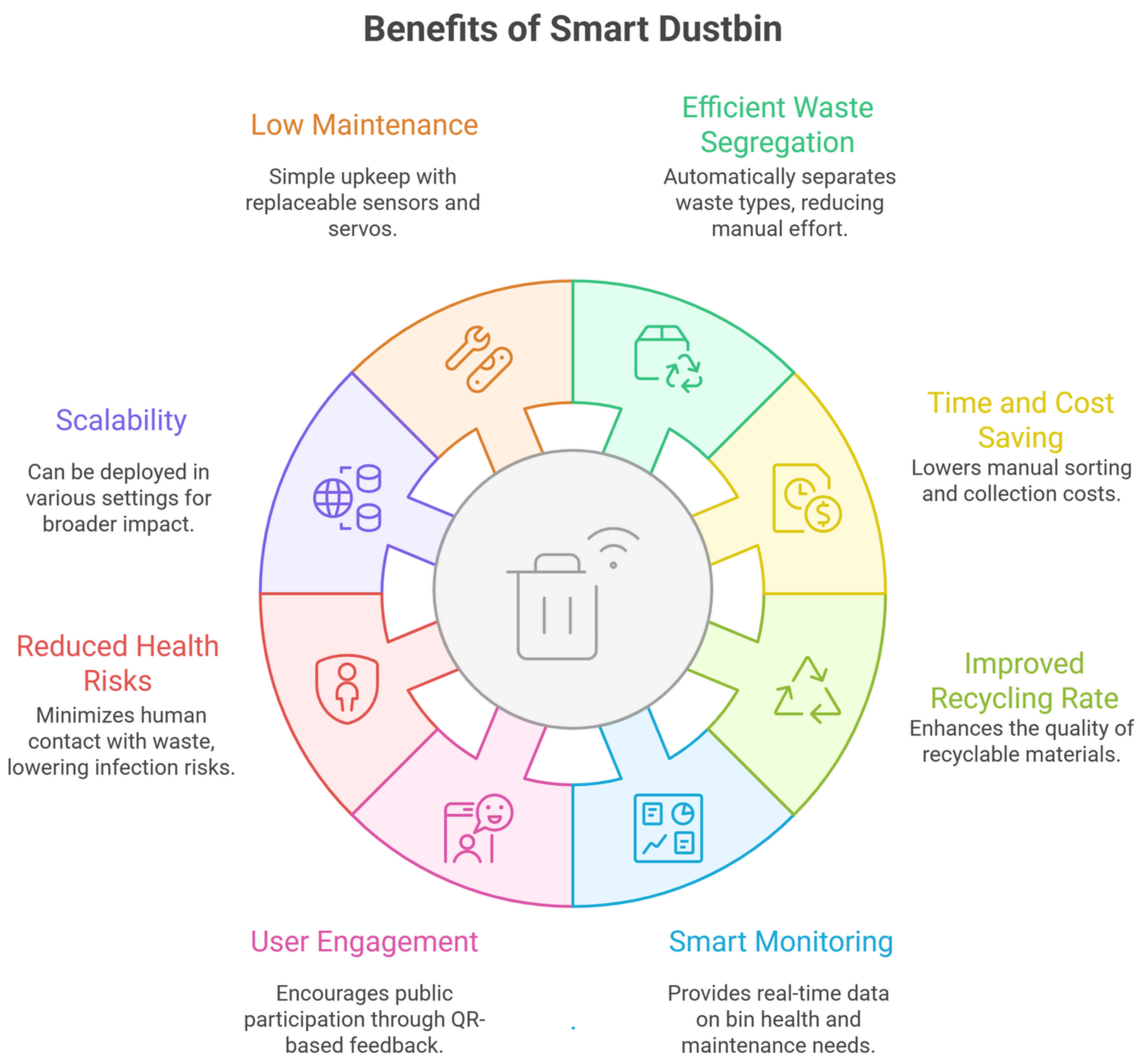
IMPACT AND BENEFITS







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RESEARCH AND REFERENCES

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Industry Outlook

- Global smart waste management market is projected to grow at ~17% CAGR (2024–2030) due to rising urbanization and smart city initiatives.
- Adoption of IoT, AI, and solar-powered bins is increasing in municipalities.
- Countries like Singapore, South Korea, and EU nations already deploy smart bins widely.
- Focus is shifting toward sustainable waste segregation to meet UN SDG (Sustainable Development Goals).

Primary Research

Surveyed sanitation workers, homemakers, and college students to gather firsthand insights.

Key Insight: The research highlighted common challenges in waste disposal and identified opportunities for smart bin adoption.

- Google Form
- Spreadsheet

References

- ScienceDirect
- PMC Article
- ResearchGate
- IIARD

Student Insights

Students are motivated to develop tech projects that promote environmental sustainability.

Smart bins combine IoT, AI, and embedded systems, providing practical learning and resume value.

College awareness campaigns can encourage adoption and proper waste disposal.

Employer Insights

Waste management companies seek low-cost, scalable bins that minimize manpower.

Employers emphasize durable, weather-proof, and vandal-resistant hardware.

Tech firms see opportunities in data analytics for waste trends and recycling.

Market Voice

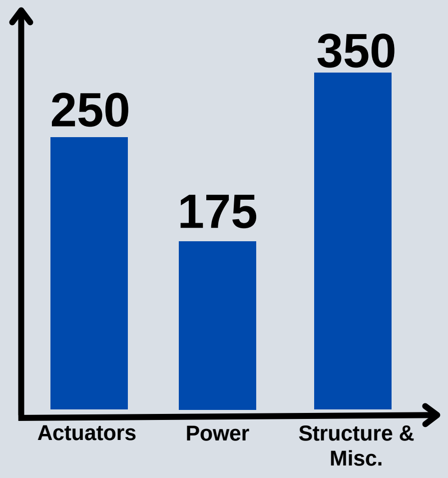
Citizens expect cleaner streets and fewer overflowing bins.

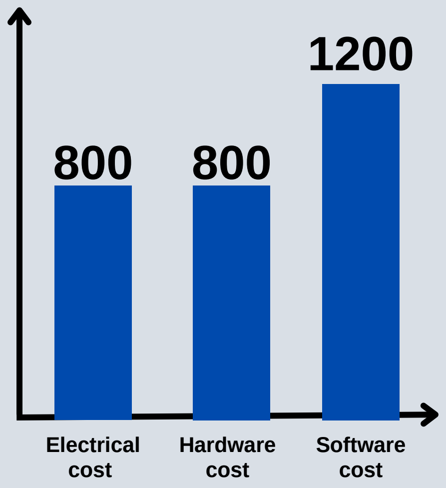
Governments promote public-private partnerships for smart waste solutions.

Investors spot opportunities in circular economy startups leveraging smart bins.

BUSINESS MODEL - SMART BIN

Cost Structure





Sensor Price Comparison

Cost Structure

- **Software Cost: ₹800**
- **Electrical Cost: ₹800**
- **Hardware Cost: ₹1,200**
- **Grand Total: ₹2,800**

Total Cost :

- Sensor Cost - ₹400
- Component Cost - ₹400
- Electrical Cost - ₹800
- Hardware Cost - ₹600

Grand Total - ₹2,800

Characteristic	Price Range (₹)
ESP32 (WiFi)	300 – 400
IR Sensor	50 – 80
Moisture Sensor	50 – 70
Ultrasonic (HC-SR04)	80 – 100
Buzzer + LEDs	40 – 60

Value Propositions

•- Automated bio vs non-bio segregation
•- Contactless IR-based lid opening
•- Real-time monitoring via dashboard
•- Reduces collection cost & improves hygiene
•- Supports sustainability and Swachh Bharat Mission

Key Partners

•- Municipal corporations & city councils
•- Facility management companies
•- Smart city solution providers
•- IoT cloud providers (AWS, Azure, etc.)
•- Recycling & waste management firms

Customer Segments

•- Municipal corporations & smart city projects
•- Corporate offices, IT parks, universities
•- Hospitals, malls, airports, railway stations
•- Residential communities & smart homes

Key Activities

•- Affordable localized solution
•- Complies with Indian waste management rules
•- Customizable dashboard for clients
•- Scalable: from single bin to city-wide network

Revenue Stream

•- One-time product sales
•- Subscription for IoT dashboard & analytics
•- Annual maintenance contracts (AMC)
•- Data monetization opportunities (future)